

Entrepreneurship

FROM DATA TO ACTIONABLE INSIGHTS

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2026



ABOUT THE INSTRUCTOR: CHUTIPORN ANUTARIYA (KAY)

- Associate Professor
 - Computer Science and Communication Technology (CSCT) Program
 - Data Science and Artificial Intelligent Applications Program
 - Faculty of Advanced Science and Technology, Asian Institute of Technology
- Area of Interests:
 - Data Science and AI, Knowledge Representation and Modeling
 - Digital Education, Digital Health, AI for Development
 - Data Analytics and Business Informatics
 - AI Ethics
 - Smart Data and Smart Technologies
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OUTLINES

- Data-driven World
- Business Analytics
- Examples
- Quiz Time
- Case Study: TELCO

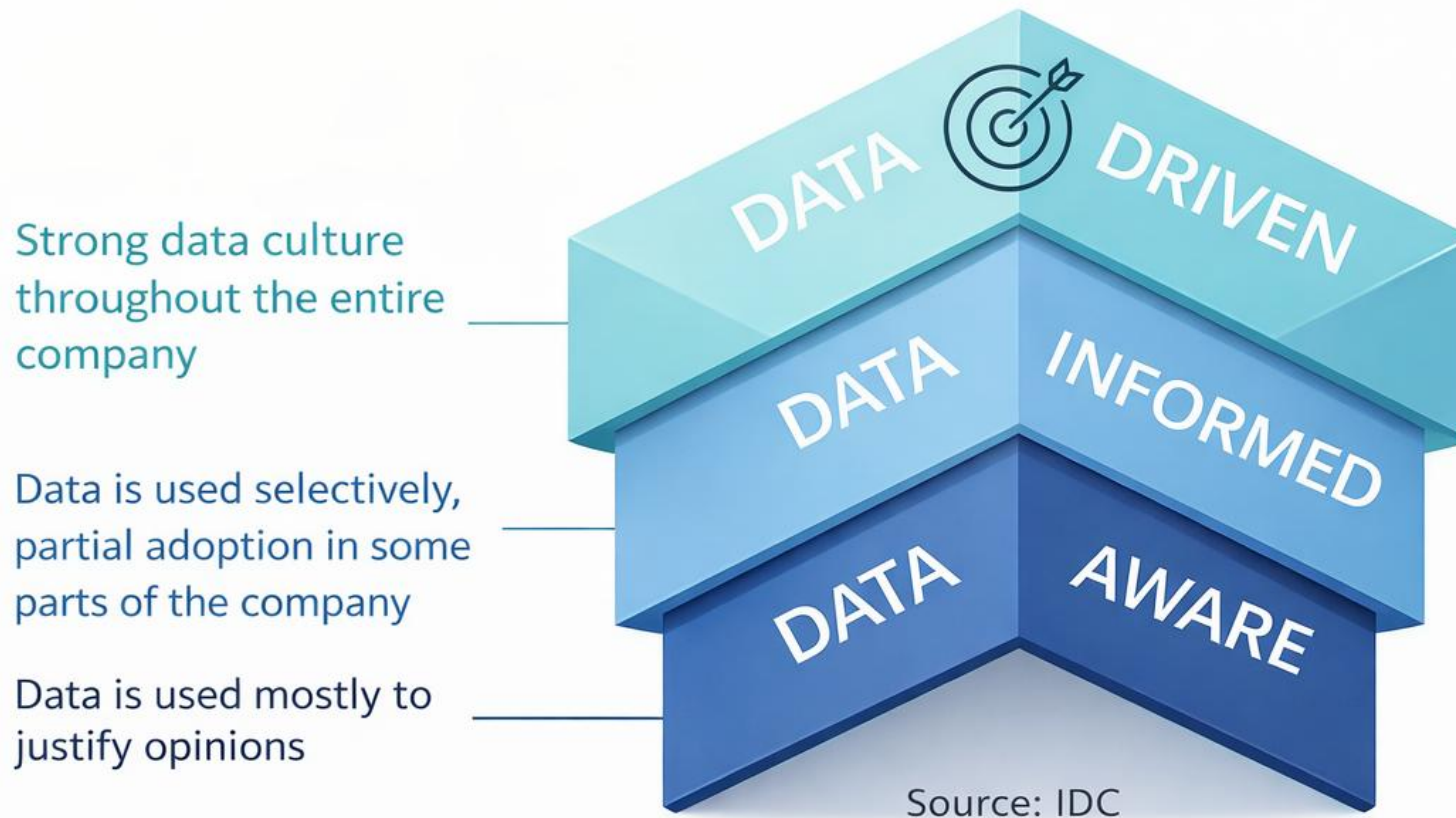


DATA-DRIVEN WORLD

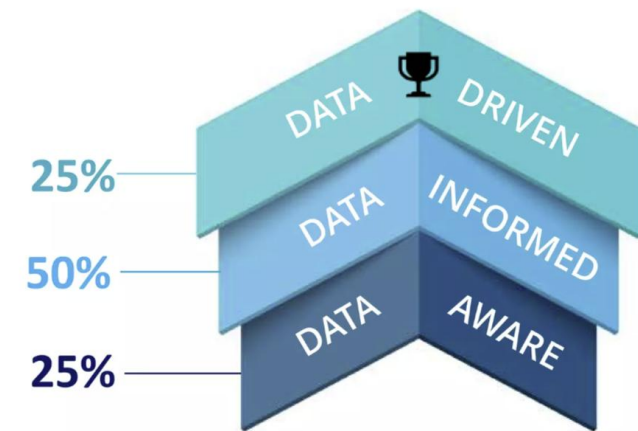
DIFFERENT LEVELS OF USING DATA WITHIN ORGANIZATIONS



There are different "levels" of relying on data



Which one are you?



Source: IDC

MOVING FROM INTUITION TO EVIDENCE-BASED PROCESS

WHAT DOES IT
MEAN TO BE A
'DATA-DRIVEN
ORGANIZATION'



What does it mean to be a “data-driven” organization?



1. Integrate Data into Decision-Making

- Prioritize data quality & accessibility.
- Embed evidence-based thinking in all workflows.
- Decisions are guided by facts, not just intuition.



2. Transform Data into Actionable Insights

- Use advanced analytics to move beyond reporting.
- Generate predictive and prescriptive insights.
- Focus on solving tangible business problems.



3. Achieve Strategic & Operational Impact

- Drive faster innovation and product development.
- Optimize processes for efficiency and cost reduction.
- Enhance customer experiences and outcomes continuously.

Data is managed and measures as an **ASSET**

1

Data is **ACCESSIBLE** and trusted

2

Data used **FREQUENTLY** in meetings

3

Data is used in **MORE THAN ONE WAY**

4



8

Data is **INFLUENCING** major decisions

7

Data is a **competitive differentiator**

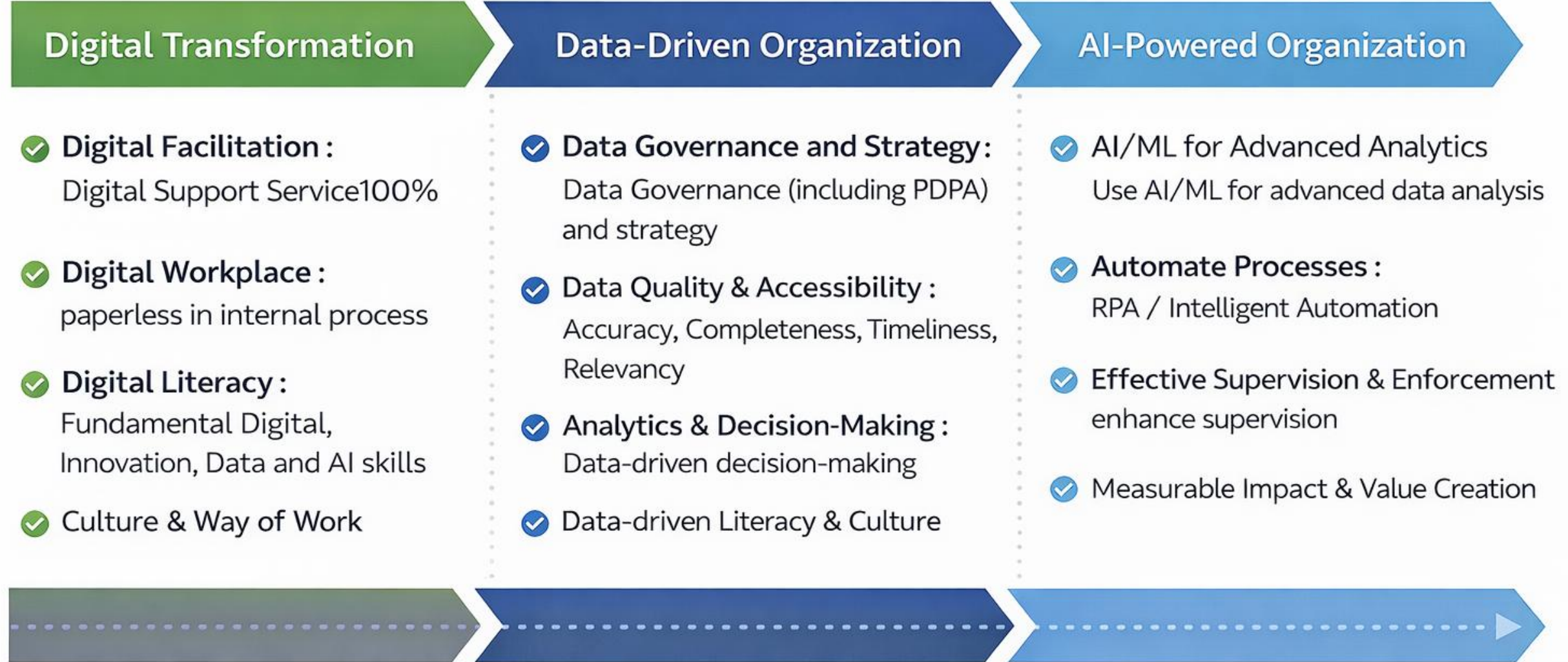
6

Culture of curiosity, innovation, experimentation

5

Data skills prioritized when hiring new employees

Transformation Journey

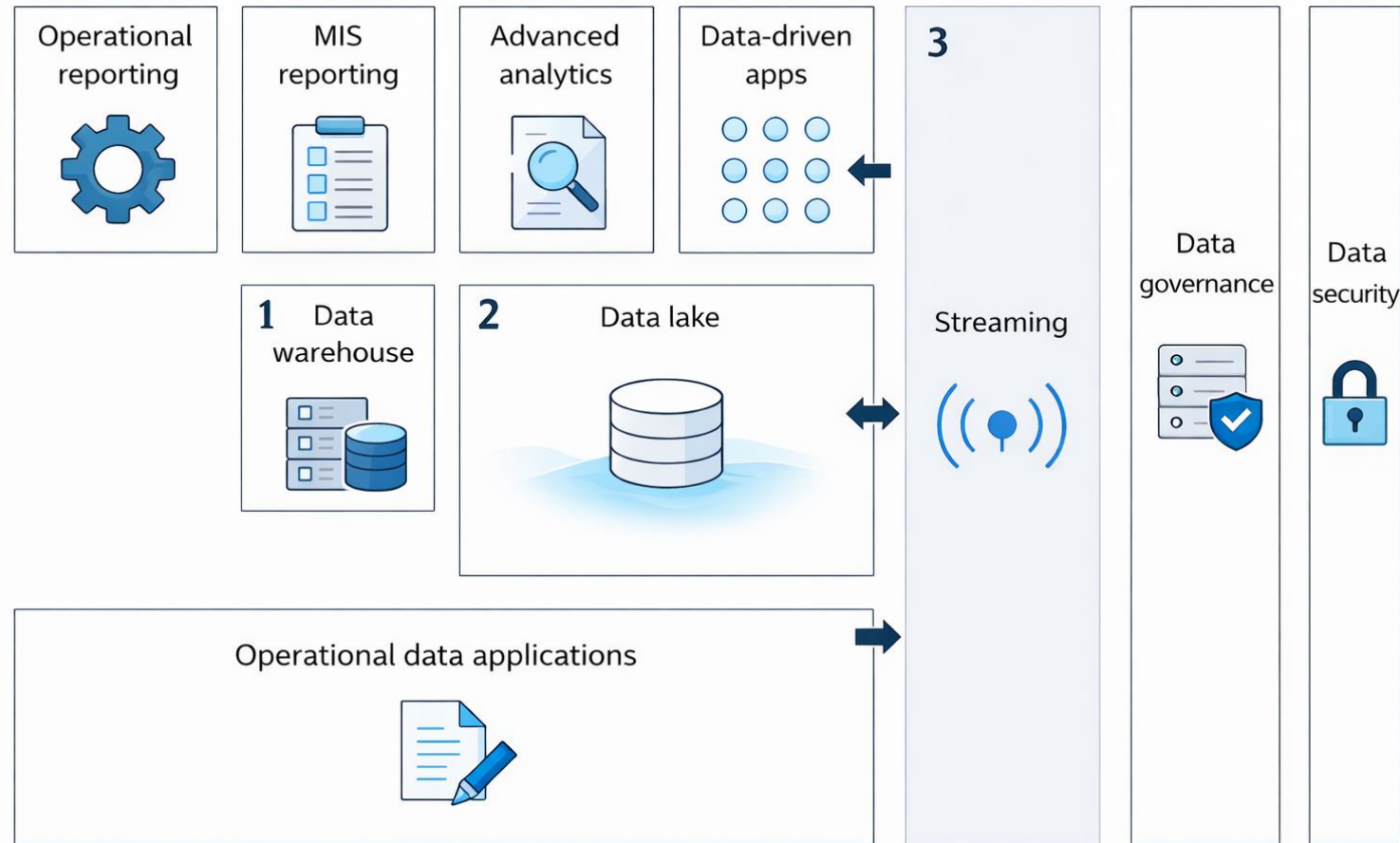


**DATA SCIENCE DOES NOT
START WITH SOPHISTICATED
MODELS. IT STARTS WITH
ACCESSIBLE, RELIABLE DATA!**

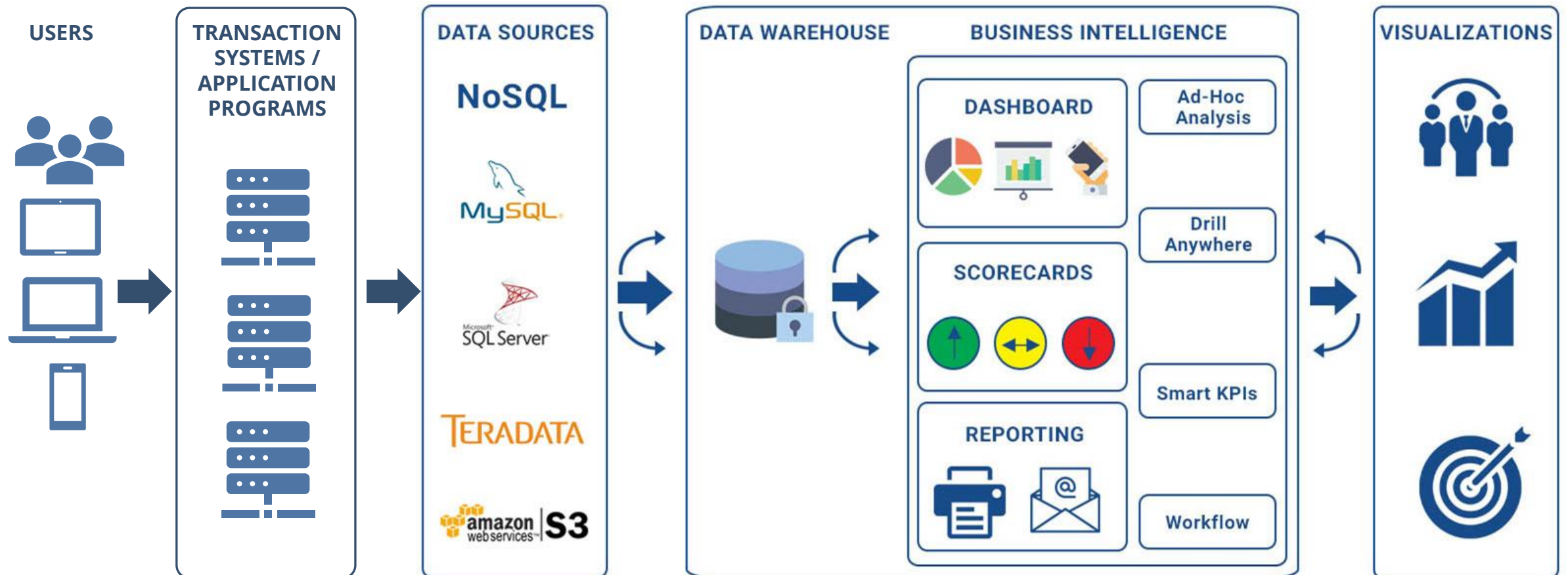
DATA PROFESSIONALS

Data Analyst	Data Architect	DB Administrator	DB Designer	Data Engineer
Data Scientist	Data Governance Specialist	Data Security Specialist	Machine Learning Engineer	BI Developer
AI Specialist	Big Data Expert	NLP Engineer	Marketing Analyst	Business Analyst

REFERENCE DATA ARCHITECTURE



Data Pipeline Architecture





BUSINESS ANALYTICS

Business Analytics Process Model



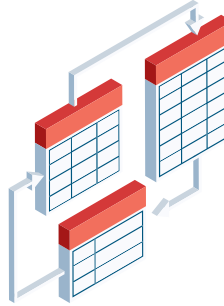
**Identify
Business
Problem**



**Identify
Data
Sources**



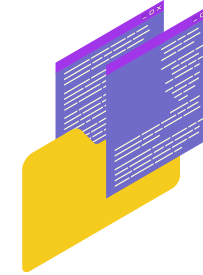
**Select
the data**



**Clean
the data**



**Transform
the data**



**Analysis
the data**



**Interpret,
Evaluate &
Deploy
the Model**

Preprocessing

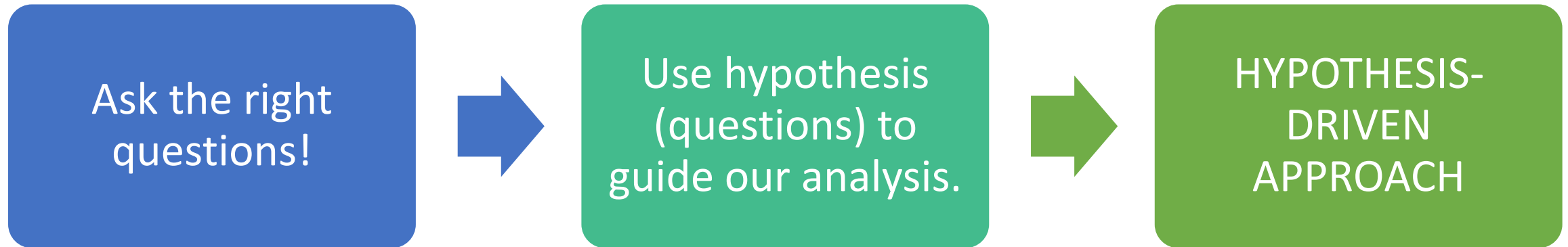
Analytics

Post processing

Business Analytics Process Model

Src: <https://blogs.sas.com/content/sgf/2019/05/14/big-data-in-business-analytics-talking-about-the-analytics-process-model/>

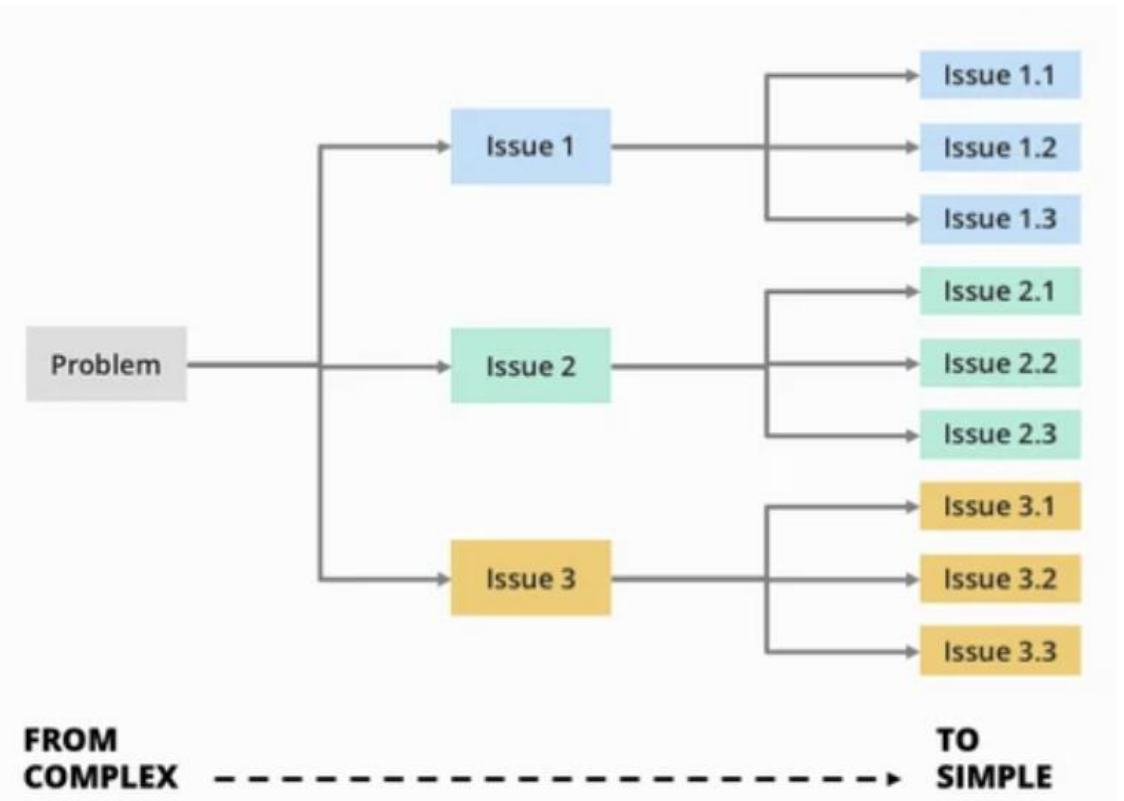
IDENTIFY BUSINESS PROBLEM



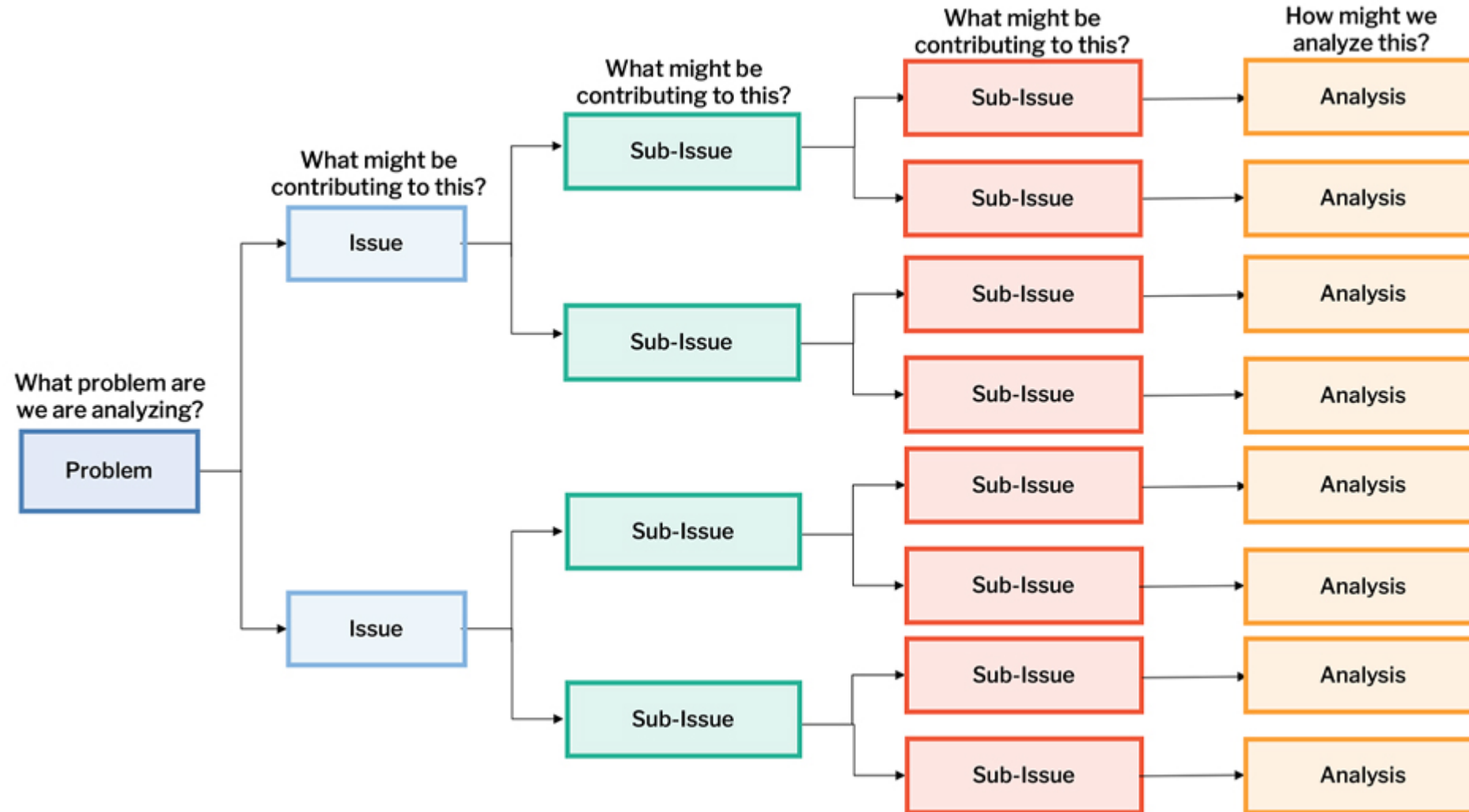
Hypothesis = a tentative assumption made in order to draw out and test its logical or empirical consequences (Merriam-Webster)

Issue (hypothesis) Tree

- A structured framework to break down and analyze complex problems or questions into smaller components.
- A visual representation of the various aspects, sub-issues, and potential solutions related to a particular problem.



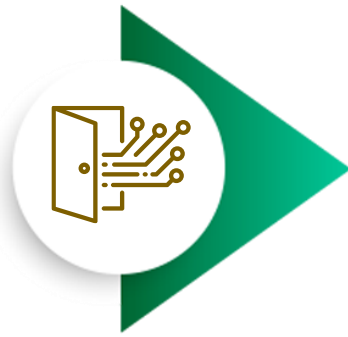
Issue Tree Template



Four Types of Data Analytics



**Descriptive
Analytics**



**Diagnostic
Analytics**



**Predictive
Analytics**



**Prescriptive
Analytics**

1

What happened

2

Why did it happen?

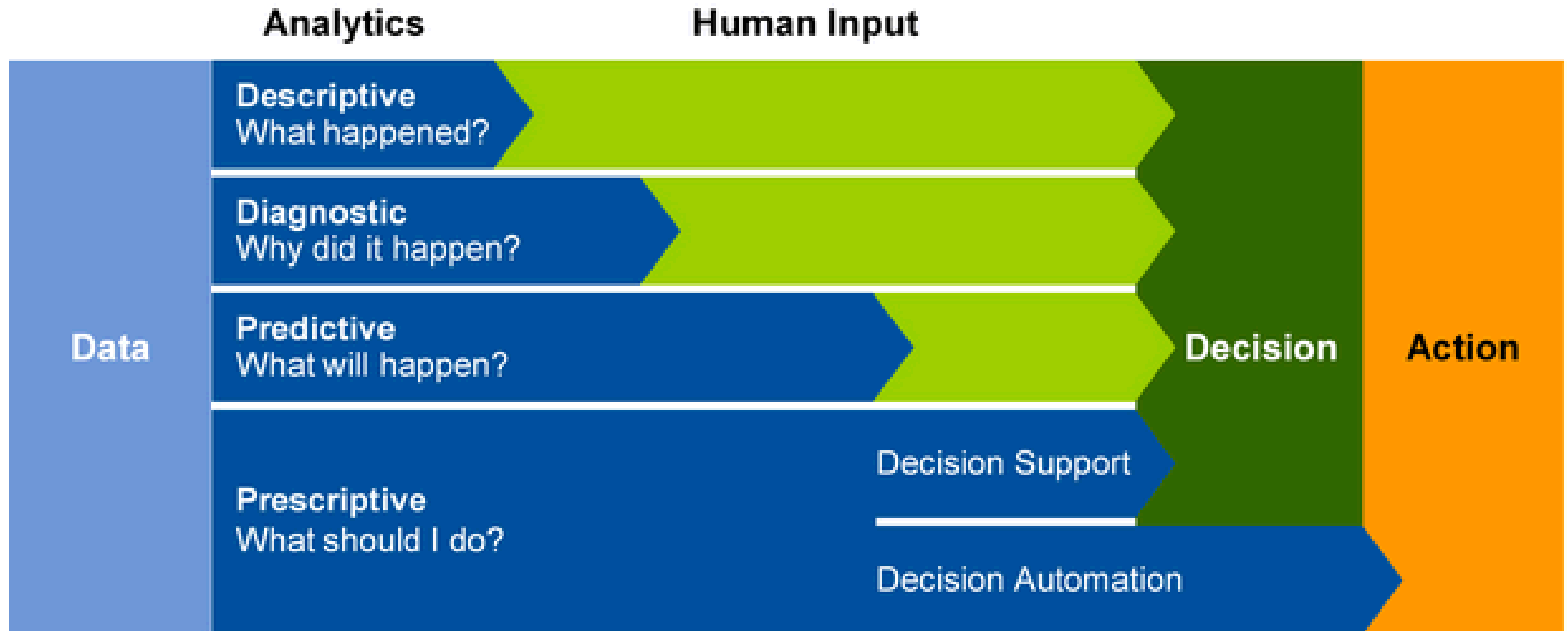
3

What will happen?

4

What should I do?

Four Types of Analytics Capability



Supporting Tools and Techniques

	DESCRIPTIVE ANALYTICS (Foundational)		DIAGNOSTIC ANALYTICS (Operational)	PREDICTIVE ANALYTICS (Insightful)	PRESCRIPTIVE ANALYTICS (Strategic)
Question	1. What happened in the past?	2. What is happening now?	3. Why did it happen and what are the relationships?	4. What will happen in the future?	5. How should we act in the future?
Process Focus	Reporting	Measuring/ Monitoring Key Performance Indicators	Trend Analysis, Situational Analysis, Root Cause, Cause and Effect & Cluster Analyses	Forecasting, Probability Assessment, Risk Management, Prediction	Scenario Based Planning, Strategy Formulation & Simulation, Option Optimization
Tools & Techniques	Static & Interactive Reports	Dashboards, Performance Scorecards	Data Mining, Modeling Statistics, Query Tools, Spreadsheets, OLAP Tools, Decision Trees	What-if Analysis, Machine Learning, Predictive Modeling, Neural Networks, Data Visualization	Discrete Choice Modeling, Linear and Non-linear Programming, Value Analysis
Retrospective			Prospective		

EXAMPLES



AI in Business Decision Making

Customer Churn Prediction

Analyzing customer behavior and engagement to proactively retain at-risk customers.

Fraud Detection

Real-time analysis of transaction data to identify and block fraudulent activity

Product Recommendation

Personalized suggestions based on browsing and purchase history to increase sales.

Dynamic Pricing

Adjusting prices in real-time based on supply, demand, seasonality, and competitor data.

Demand Forecasting

Optimizing inventory, production, and logistics to meet future customer demand efficiently.

AI Techniques in Business Analytics

Classification

Used to predict categories or classes.

Example: Predicting whether a customer will churn or stay.

Regression

Used to predict numerical values.

Example: Forecasting sales, revenue, or demand.

Clustering

Used to group customers based on similar behavior or characteristics.

Example: Customer segmentation for marketing.

Recommendation System

Used to predict suggested products or services.

Example: Netflix movie recommendations or Amazon product suggestions.

Forecasting

Used to predict future trends based on historical data and patterns.

Example: Forecasting customer demand, network traffic, or monthly subscription growth.

Anomaly Detection

Used to identify unusual patterns, outliers, or suspicious activities in data.

Example: Detecting fraudulent transactions or abnormal service usage.

Association Rule Mining

Used to discover relationships between products, services, or customer behaviors.

Example: Finding that customers who buy a mobile data package often also subscribe to a streaming service.

Decision Trees

Used to make decisions by following a tree-like structure of conditions and outcomes.

Example: Deciding whether a customer should receive a discount based on tenure, usage, and churn risk.

Optimization

Used to find the best possible action or decision under certain constraints.

Example: Selecting the best retention offer that maximizes customer satisfaction while controlling cost.

Natural Language Processing

Used to analyze, understand, and generate human language from text or speech.

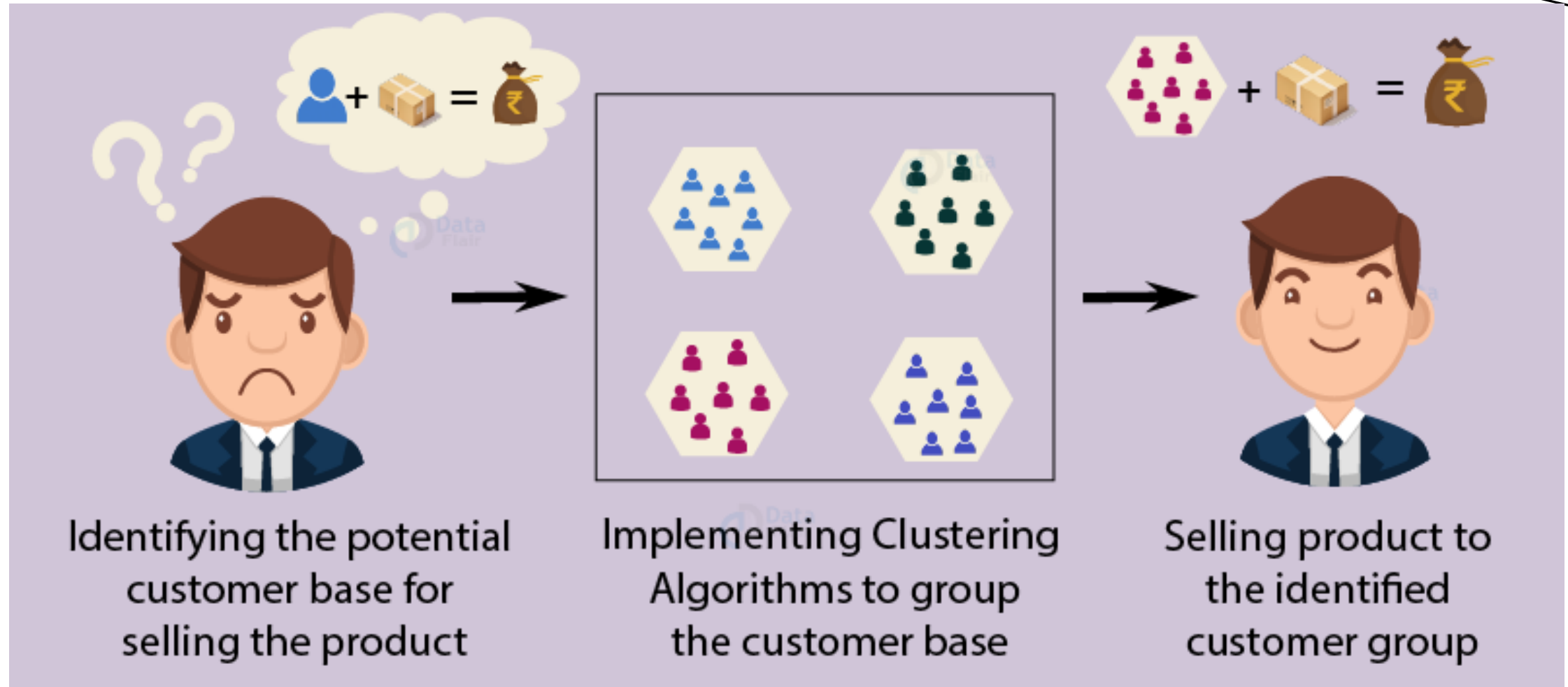
Example: Extracting key topics from customer service chats or support tickets.

Sentiment Analysis

Used to identify customer opinions, emotions, or satisfaction levels from text.

Example: Detecting whether customer reviews, social media posts, or survey comments are positive, negative, or neutral.

Customer Segmentation



Sentiment Analysis



Analyzing the Market
Response to a Product



Managing & Monitoring
Business Reputation



Social Media
Monitoring

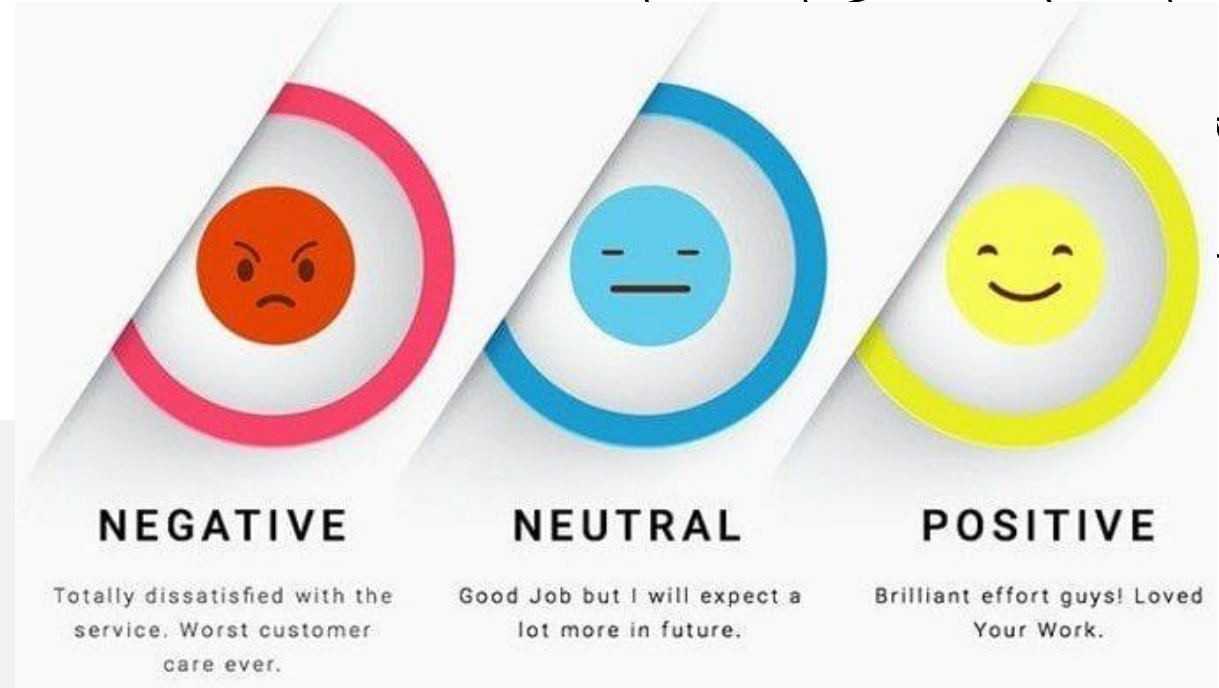
**5 Key Business
Applications of
Sentiment Analysis**



Get to Know the Voice
of Your Customers (VoCs)



Analyzing Market &
Competitors



WHERE AI Creates Values

01

Revenue Growth

- Identify upsell opportunities, optimize pricing strategies, and predict customer lifetime value.

02

Cost Reduction

- Streamline operations, reduce waste, and optimize resource allocation through intelligent automation.

03

Risk Reduction

- Predict and prevent fraud, detect anomalies, and assess credit risk with greater accuracy.

04

Productivity Improvement

- Automate routine tasks, prioritize high-value activities, and enhance decision-making speed.

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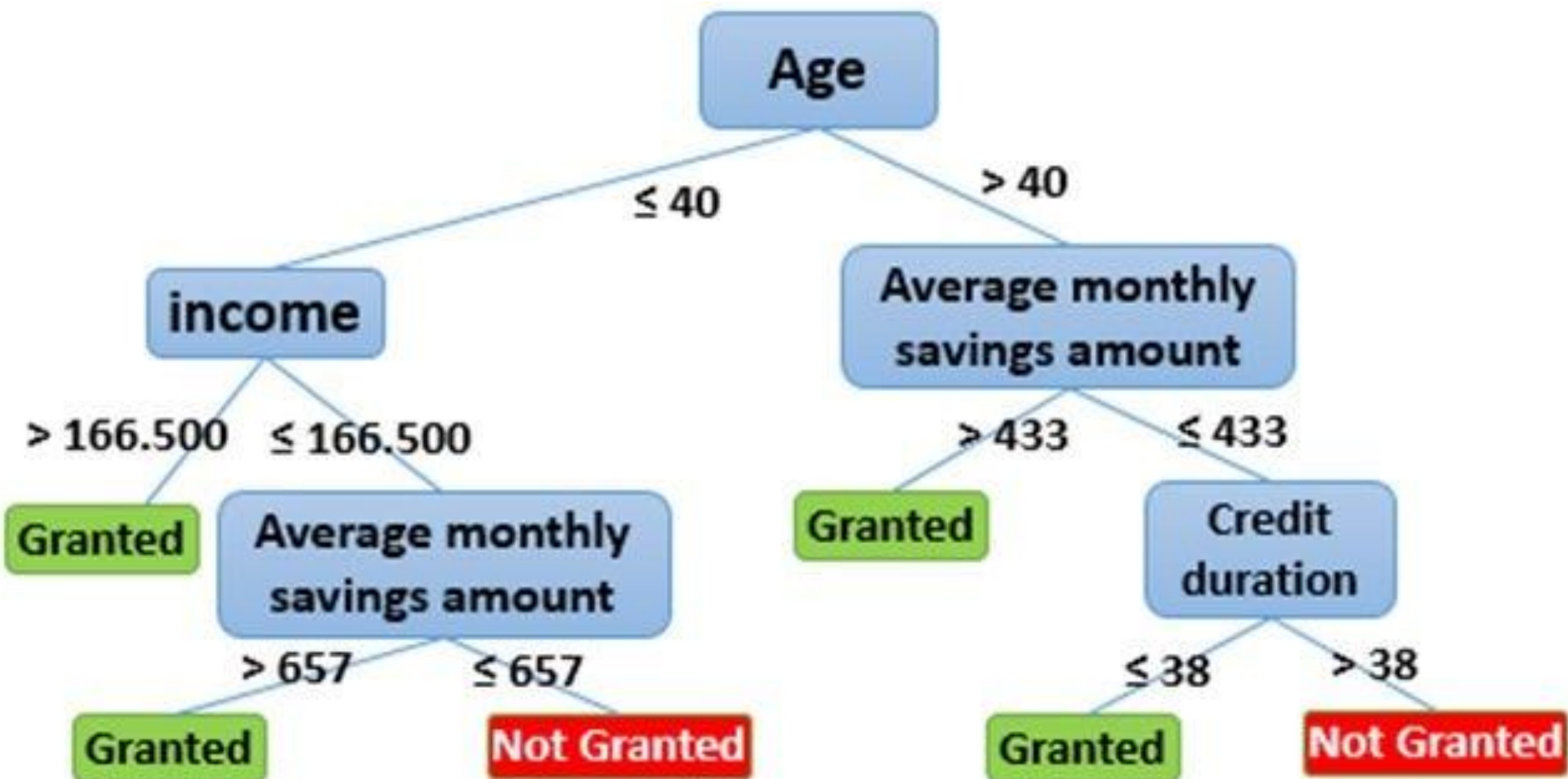
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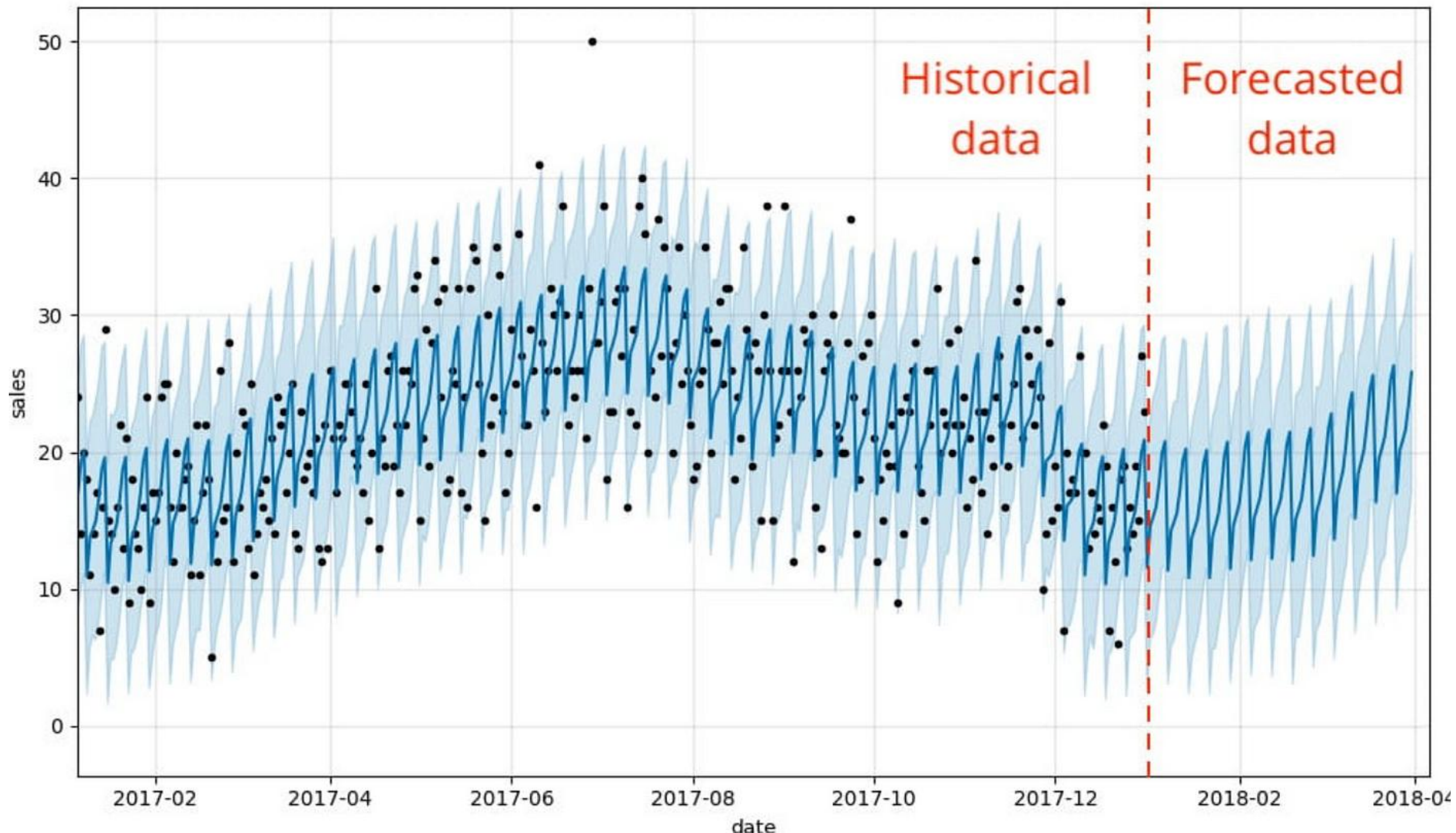
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Top Picks for Veronica



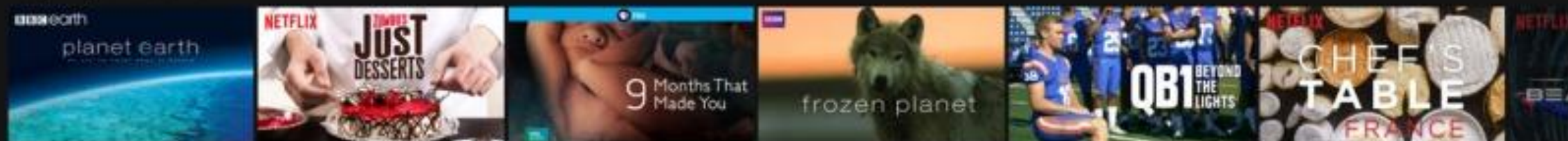
New Releases



Because you watched 13 Reasons Why



Inspiring TV Shows





Shopping cart



Placeholder text for the shopping cart item.

People also bought



Recommended for you

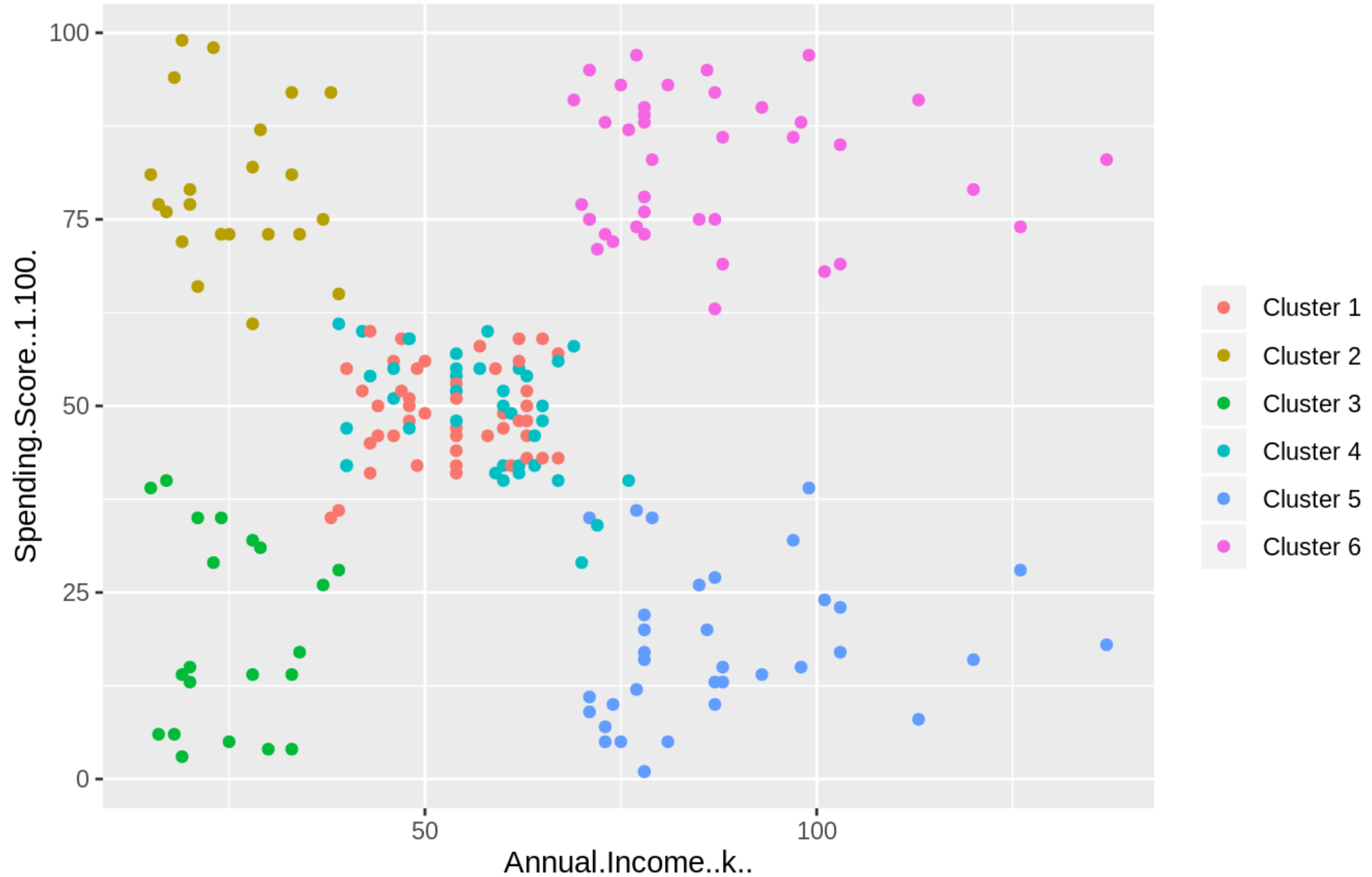


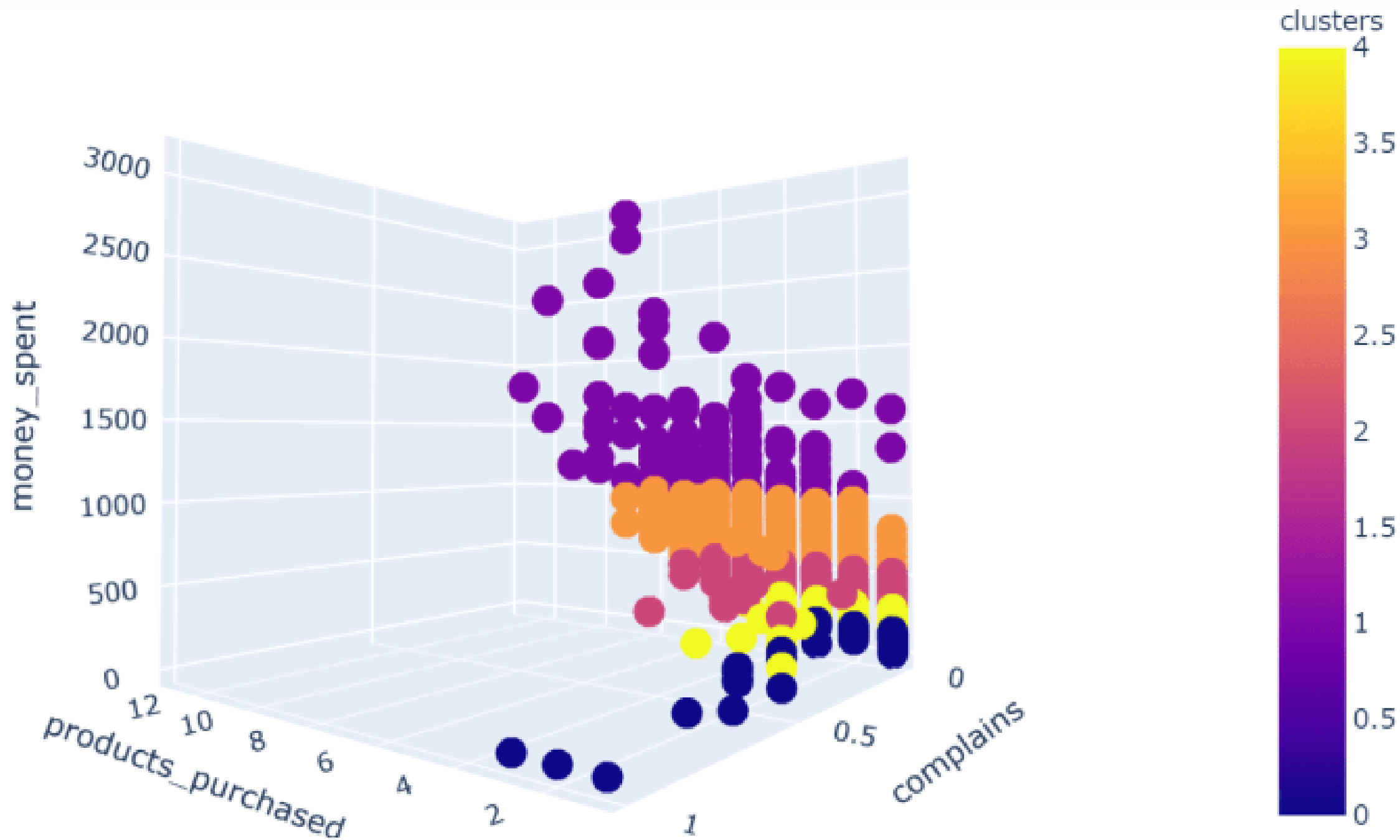
You might like



Segments of Mall Customers

Using K-means Clustering

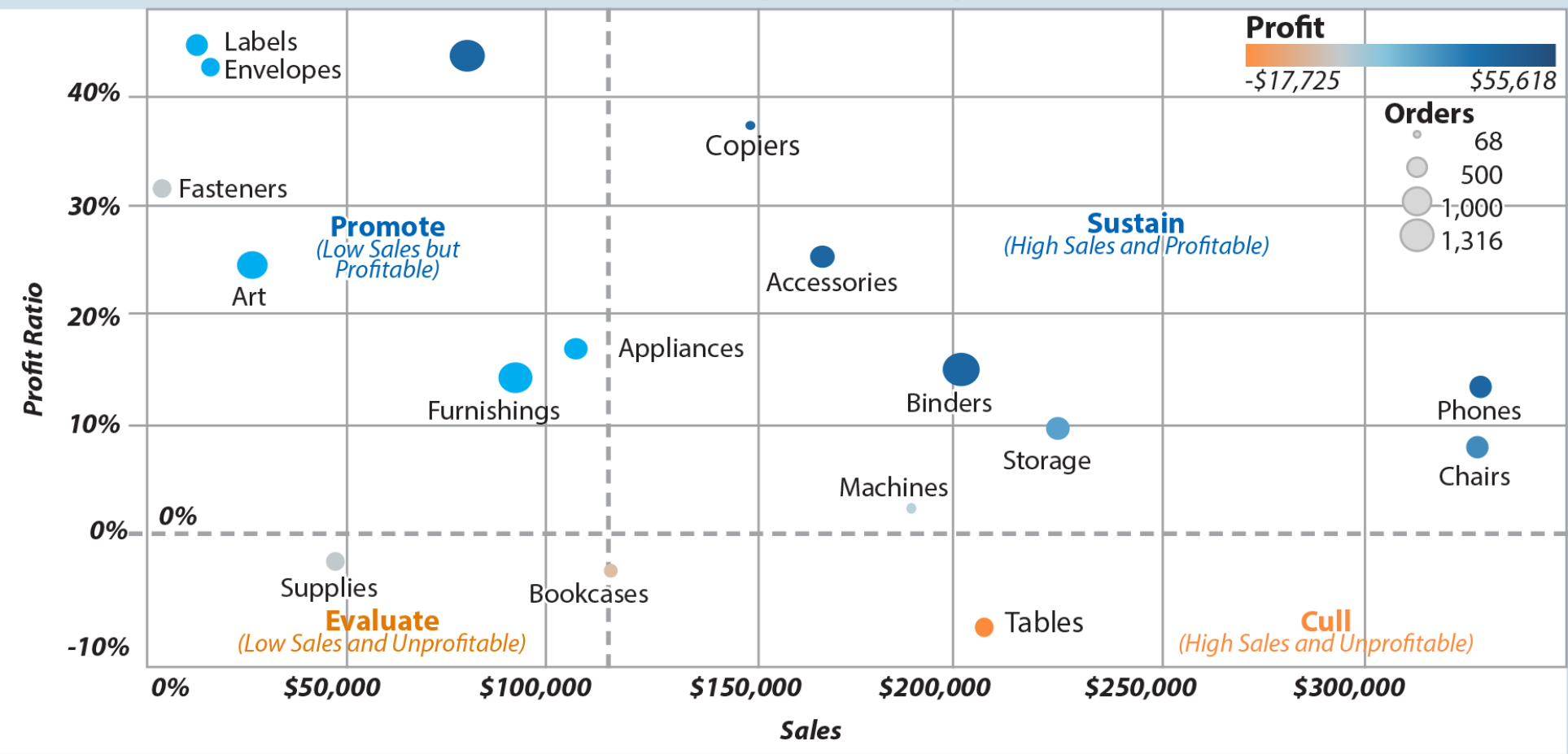




Product Profitability Analysis – What Products are Performing Well?

Profit Ratio vs. Sales:

Click on a product Group to filter



Very disappointed with the support...



Patt Marlowe [Director of IT, Green Bay Packers]

To: support@technology.com

Hello support,

We are not happy with the progress on this case. Your previous suggestion to follow the KB article didn't help. Maybe it would be useful if your product had a section for tutorials.

This problem was a critical failure at the time and we expected a timely RCA. This is complete bulls&@# - I don't understand the direction you are trying to analyze the case.

I also tried installing the latest patch (v3.2) on my AWS EC2 production instance, but with no luck. We need it to make progress on a decision today using this - can you help me with this in the next hour? If you cannot resolve this I need to speak with your manager today.

Also we have been working with our Sales Rep re: renewal to happen next month. Because of these issues our CIO considering to take our business elsewhere.



Mia Gonzales [Senior support engineer at Technology Inc.]

To: pat.marlowe@gbpackers.com

Thank you for your patience. We are receiving high volume of cases, so it might be slight delay from our side. I understand that you are upset, we are still working on this issue, and I don't have any updates for you as of right now.

SupportLogic

Customer Signals

Sentiment	Impact
Lack of progress	Critical issue
Frustration	Urgency
Profanity	Production Issue
Feedback	Churn Risk
Documentation Gap	Escalation Request
Confusion	Revenue Impact

Keywords

RCA
Patch (v3.2)
AWS EC2
CIO

Agent signals

Politeness
Empathy
Lack of progress

Assessment

80	Attention Score	Poor	Agent Interaction
30	Sentiment Score	Account Health	

Predictions

Likely to escalate
Churn risk

Terrible Service - I ordered a a brand new computer from the web store. I'm furious to find out that the tracking number doesn't work and my order might be lost. Please respond ASAP.

Order Issue

Negative

Urgent





Sprout Coffee Co.
Listening



Listening Home



Active Topics



Featured Topics



+ New Topic



Java House



Holiday Lattes



Coffee Cabin



Talk #SproutLove



Sprout Coffee Care Chicago



Sprout Coffee Care



Sprout Coffee



Archived Topics



Twitter Search



Sprout Coffee

Data available from Dec 1, 2018 – Present

1/1/20

→ Messages: Saturday, January 7, 2023



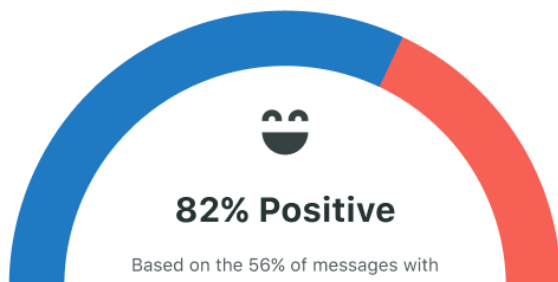
Performance

Conversation

Demographics

Sentiment Summary

Understand how users feel about this topic and see how positive and negative sentiment changed since the



Sentiment Trends

Net Sentiment Score

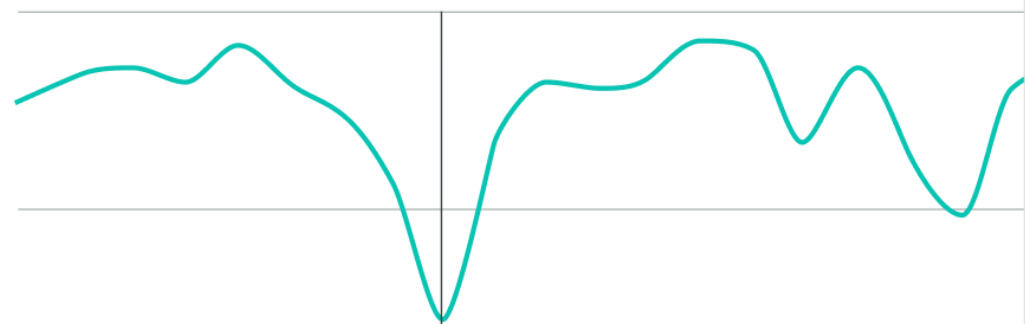
38% difference between positive and ne

Net Sentiment Trend

Your net sentiment score decreased by

Sentiment Trends

View the positive and negative sentiment changes over time for this reporting period.



Spike Alert Summary ⓘ

Spike Alert detected at 8AM. Top keyword appearing during this spike is **Matcha**

Potential Impressions ▾

Message

Jan 7, 2023 8:23 am



Minnie Watkins 100k +
@minniemakes

Sprout Coffee is typically has the best coffee ever and is my go-to. Not sure why but today there was something off about the matcha latte. I wonder if t...

[Show More](#)

Matcha

Latte



304 Engagements



Message

Jan 7, 2023 9:00 am



Hannah Scott
@hannahscott

[@mysproutcoffee](#) did your matcha recipe change? My drink this morning was flavorless.

543 Engagements

CASE STUDY: TELCO



Telecom Customer Retention & Service Usage

Descriptive Analytics: Understanding Customer Usage Patterns

- Analyze historical data:
 - Call usage,
 - Mobile data consumption,
 - Subscription plans,
 - Regional customer distribution.
- Identify:
Most popular plans,
High-usage customer segments,
Regions with highest customer base.

Diagnostic Analytics: Investigating Customer Churn

- Investigate causes:
Why customer churn increased in a specific plan?
- Analysis shows:
 - High data usage customers experience slower network speeds
 - Customers switching to competitors with better pricing.
- Investigate network quality and pricing strategies.

Telecom Customer Retention & Service Usage

Predictive Analytics: Predicting Customer Churn

- Predict:
Which customers are likely to leave the telecom service?
 - Contract type
 - Monthly charges
 - Customer tenure
 - Service usage patterns.
- Build predictive churn model:
 - Estimate probability of customers leaving in next few months, Prioritize high-risk customers.

Prescriptive Analytics: Optimizing Customer Retention

- Optimize:
Customer retention strategies using churn predictions.
- Recommend actions:
 - Offering loyalty discounts
 - Upgrading data packages
 - Providing targeted promotional offers.
- Retain valuable customers and reduce churn.



SUMMARY & KEY TAKE AWAYS

